Changes in serum concentrations of polychlorinated biphenyls (PCBs), hydroxylated PCB metabolites and pentachlorophenol during pregnancy.

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We studied pregnancy-related changes in serum concentrations of five polychlorinated biphenyls (PCBs, CB 118, CB 138, CB 153, CB 156, CB 180), three hydroxylated PCB metabolites (4-OH-CB107, 4-OH-CB146, 4-OH-CB187), and pentachlorophenol (PCP). Median serum lipid content increased 2-fold between early (weeks 9-13) and late pregnancy (weeks 35-36) (N=10), whereas median PCB levels in serum lipids decreased 20-46%, suggesting a dilution of PCB concentrations in serum lipids. Nevertheless, strong positive intra-individual correlations (Spearman's r=0.61-0.99) were seen for PCBs during the whole study period. Thus, if samples have been collected within the same relative narrow time window during pregnancy, PCB results from one single sampling occasion can be used in assessment of relative differences in body burdens during the whole pregnancy period. Concentrations of OH-PCBs in blood serum tended to decline as pregnancy progressed, although among some women the concentrations increased at the end of pregnancy. Positive intra-individual correlations (r=0.66-0.99) between OH-PCB concentrations were observed during the first and second trimester, whereas correlations with third trimester concentrations were more diverging (r=-0.70-0.85). No decline in PCP concentrations was observed during pregnancy and no significant correlations were found between concentrations at different sampling periods. Our results suggest that for both OH-PCBs and PCP, sampling has to be more specifically timed depending on the time period during pregnancy that is of interest. The differences in patterns of intra- and inter-individual variability of the studied compounds may be due to a combination of factors, including lipid solubility, persistence of the compounds, distribution in blood, metabolic formation, and pregnancy-related changes in body composition and physiological processes.